

RESEARCH METHODOLOGY

CIE-1

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ANOVA Analysis Report on Netflix Dataset

1. Introduction

Netflix is one of the leading online streaming platforms that offers a large collection of movies and TV shows across different genres, durations, and audience ratings. Analyzing such data helps in understanding patterns and trends in content available on the platform.

To examine this, Analysis of Variance (ANOVA) is used. ANOVA is a statistical technique that helps determine whether there are significant differences between the means of multiple groups. In this report, ANOVA is applied to analyze whether the average movie duration differs significantly across different rating categories in the Netflix dataset.

2. Objective

The objective of this analysis is to determine whether there is a statistically significant difference in the average duration of movies across different rating categories using One-Way ANOVA.

3. Dataset Description

The dataset contains Netflix titles with attributes such as:

- Type (Movie/TV Show)
- Title
- Country
- Release Year
- Rating
- Duration
- Genre (listed_in)

For this analysis:

- Only **Movies** were selected.
- Duration was converted into **numeric minutes**.
- Ratings with sufficient data points were considered.

4. Problem Statement

The Netflix dataset contains movies with varying durations and different audience rating categories. However, it is not clear whether the difference in movie durations across these rating categories is statistically significant or just due to random variation.

5. Hypothesis

- **Null Hypothesis (H_0):**
There is **no significant difference** in the mean movie duration across different rating categories.
- **Alternative Hypothesis (H_1):**
There **is a significant difference** in the mean movie duration across rating categories.

6. Methodology

- Extracted movie records from dataset.

- Converted duration into numeric format (minutes).
- Grouped movies based on their **ratings** (e.g., TV-MA, PG, R, etc.).
- Applied **One-Way ANOVA test** using these groups.

7. ANOVA Results

- **F-statistic:** 126.58
- **p-value:** 5.30×10^{-241}

8. Interpretation

- The **p-value is extremely small ($<< 0.05$)**.
- Therefore, we **reject the null hypothesis**.

This indicates that:

There is a **statistically significant difference** in average movie duration among different rating categories.

9. Implementation (Python Code)

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('netflix_titles.csv')

movies = df[df['type'] == 'Movie'].copy()

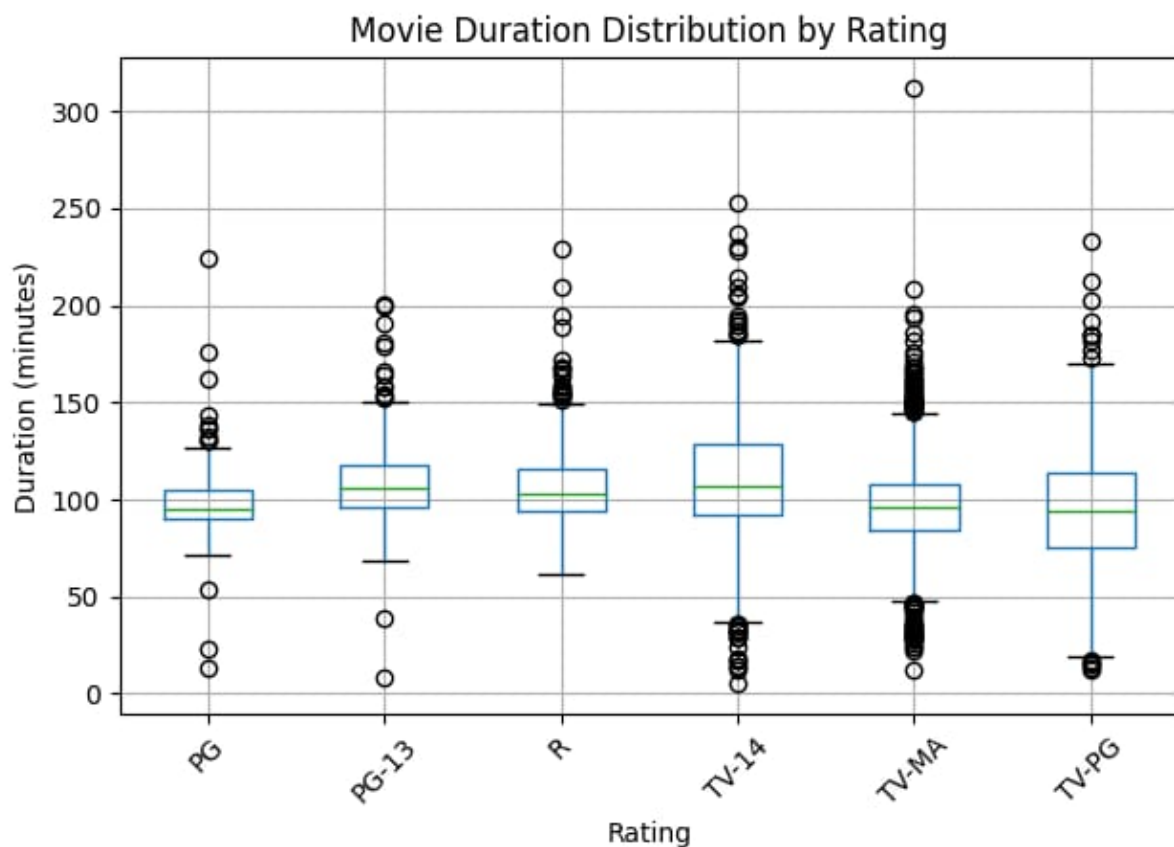
movies['duration'] = movies['duration'].str.replace(' min', '')
movies['duration'] = pd.to_numeric(movies['duration'], errors='coerce')

movies = movies.dropna(subset=['duration', 'rating'])
top_ratings = movies['rating'].value_counts().head(6).index
filtered = movies[movies['rating'].isin(top_ratings)]
```

10. Visualization

1. Boxplot: Duration vs Rating

```
plt.figure()
filtered.boxplot(column='duration', by='rating')
plt.title('Movie Duration Distribution by Rating')
plt.suptitle("")
plt.xlabel('Rating')
plt.ylabel('Duration (minutes)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

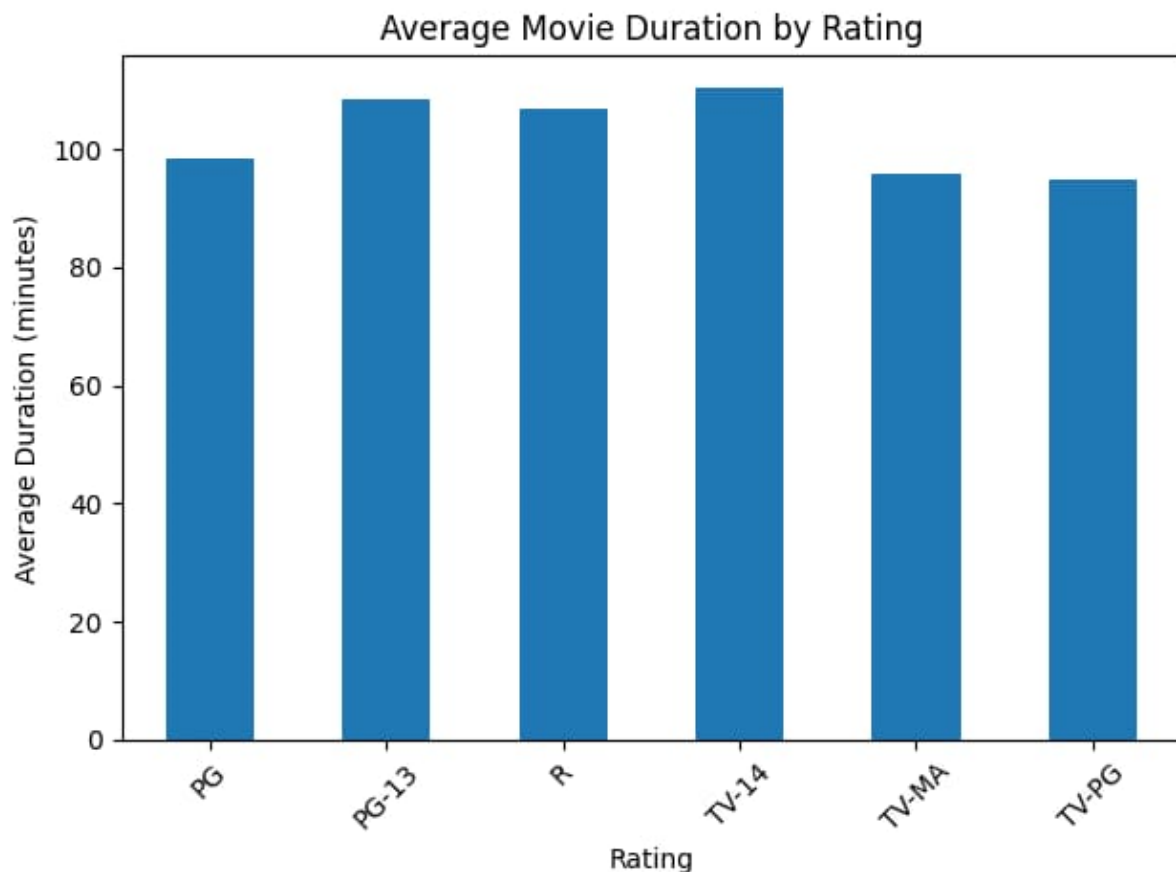


Bar Chart: Average Duration by Rating

```
mean_duration = filtered.groupby('rating')['duration'].mean()
```

```
plt.figure()
```

```
mean_duration.plot(kind='bar')
plt.title('Average Movie Duration by Rating')
plt.xlabel('Rating')
plt.ylabel('Average Duration (minutes)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



8. Conclusion

The ANOVA test confirms that **movie duration varies significantly across rating groups**. This insight can help:

- Content creators plan movie length based on target audience.
- Streaming platforms optimize recommendations.